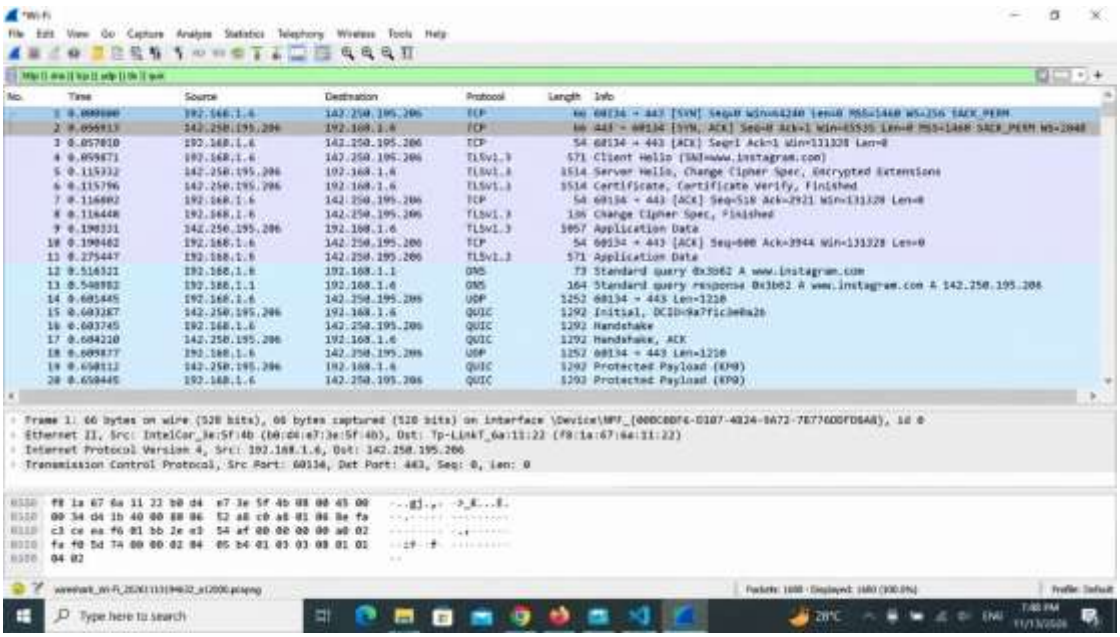
Day 1: Introduction to Networking and the OSI ModelTheory (1 hour): What is Networking?Overview of the OSI Model: Layers and their functions. Practical (1 hour):Use a network sniffer (e.g., Wireshark) to observe traffic and identify layers. Task: Create a diagram of the OSI Model with examples for each layer.

Tasks

1.Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

ANS:

1. Download Wireshark from the official website: https://www.wireshark.org/download.html
2. Install Wireshark on your laptop.
3. Open Wireshark.
4. Select your active Wi-Fi or Ethernet network interface.
5. Click **Start Capture**.
6. Open Google Chrome and visit www.instagram.com.
7. Wait for a few seconds.
8. Return to Wireshark and click **Stop Capture**.
9. Observe the Protocol column in the packet list.



No.	Protocol	Purpose
1	DNS	Converts domain names into IP addresses.
2	TCP	Provides reliable data transmission.
3	TLS	Secures communication through encryption.
4	UDP	Transfers data without establishing a connection.
5	QUIC	Supports fast and secure internet communication.
6	HTTP	Transfers web page information.
7	IPv4 / IPv6	Provides IP addressing and packet delivery.
8	ARP	Finds the MAC address associated with an IPv4 address.
9	ICMP	Used for network diagnostic messages.

2.Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

ANS:

1. Packet Identification

Based on the sample Wireshark capture, here are three example packets:

Packet No.	Protocol	Packet Information	OSI Layer
1	DNS	Resolves www.instagram.com into an IP address	Layer 7 – Application
2	TCP	Establishes a connection with the server using SYN	Layer 4 – Transport
3	IPv4	Transfers packets between source and destination IP addresses	Layer 3 – Network

2. Explanation

Packet 1 – DNS (Application Layer)

DNS converts a website name, such as instagram.com, into an IP address so the computer can locate the website server.

Packet 2 – TCP (Transport Layer)

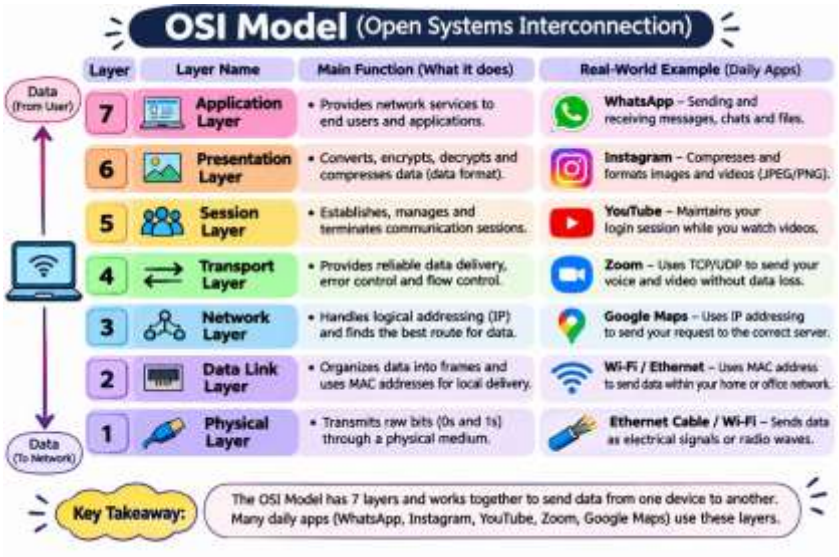
TCP establishes a reliable connection between my laptop and the website server and ensures data is delivered correctly.

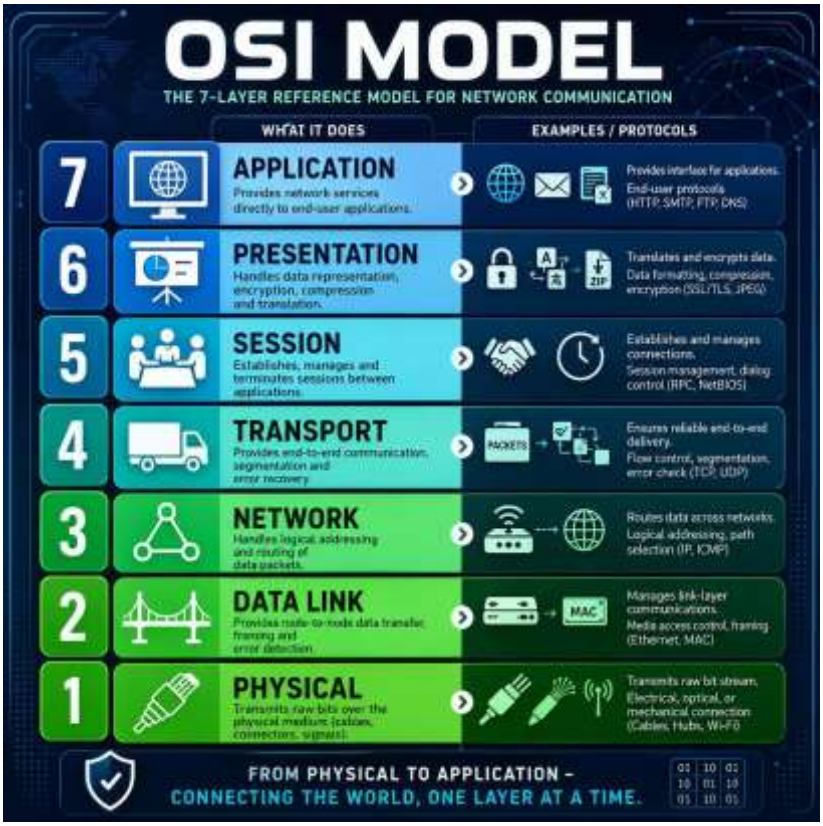
Packet 3 – IPv4 (Network Layer)

IPv4 handles logical addressing and helps transfer packets from the source device to the destination device.

3.Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

ANS:





OSI Layer	Daily-life example
7. Application	WhatsApp – Sending messages
6. Presentation	Instagram – Image formats and encryption
5. Session	YouTube – Maintaining login sessions
4. Transport	Zoom – TCP/UDP communication
3. Network	Google Maps – IP addressing
2. Data Link	Wi-Fi / Ethernet – MAC addresses
1. Physical	Ethernet cable / Wi-Fi signals

4.Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.

ANS:

1. Application Layer (Layer 7)

I open WhatsApp, type the message "Hello" and press the Send button. The application layer provides the interface and network services needed to send my message.

2. Presentation Layer (Layer 6)

The message is prepared in a suitable data format and protected using encryption. WhatsApp uses end-to-end encryption to protect personal messages.

3. Transport Layer (Layer 4)

The data is handled by transport protocols such as TCP. This layer helps deliver data between the communicating devices or services.

4. Network Layer (Layer 3)

The IP address helps identify the destination, and routers forward packets through different networks towards the destination.

5. Data Link Layer (Layer 2)

The data is organized into frames and transmitted over the local network using MAC addresses, such as through Wi-Fi.

6. Physical Layer (Layer 1)

The data is transmitted as electrical signals, radio waves or light signals through the network medium.

Message Receiving Process

When the message reaches the receiver's phone, the data is processed through the networking layers in reverse order. WhatsApp decrypts the message on the receiver's device, and the receiver can read "Hello".

